Do Consumers in Developed and Developing Countries Exhibit Different Spending Adjustment Patterns During Periods of Inflation?

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How do inflation and other economic variables affect consumption differently in developed and developing countries, and do these effects occur with a time lag?

Data Source:
World Bank (global macroeconomic indicators)

Time Period:
1991 – 2019

Countries Included
Developed: Australia, United States ,France
Developing: Thailand, Philippines, Malaysia

Variables Used
Independent Variables:

I. Inflation
II. GDP per capita
III. Unemployment rate
IV. Exchange rate
V. Gross saving

Target Variable

I. Household Final Consumption

Modeling Approach

Multiple Linear Regression
→ Used to examine the relationship between economic variables and consumption
Lagged Models (Lag 1 & Lag 2)
→ Incorporate past values of inflation and other macroeconomic variables
→ Capture delayed effects on consumption
→ Improve model performance and reduce autocorrelation

Clustering Analysis

K-Means Clustering
→ Applied to identify natural groupings in the data
→ Used to examine whether countries cluster into developed and developing groups based on economic patterns

Summary Statistics

Variable	Mean	Std Dev	Min	Max	Range
Consumption	3.78	2.97	-10.24	13.03	23.26
Inflation	2.88	2.27	-0.9	19.26	20.16
GDP per Capita	24529.47	21044.15	1620.36	60750.99	59130.63
Unemployment rate	5.08	3.02	0.25	12.59	12.34
Exchange rate	14.02	17.93	0.68	56.04	55.36
Gross Saving	26.49	6.77	13.89	39.85	25.96

Correlation with Consumption – Developed Countries

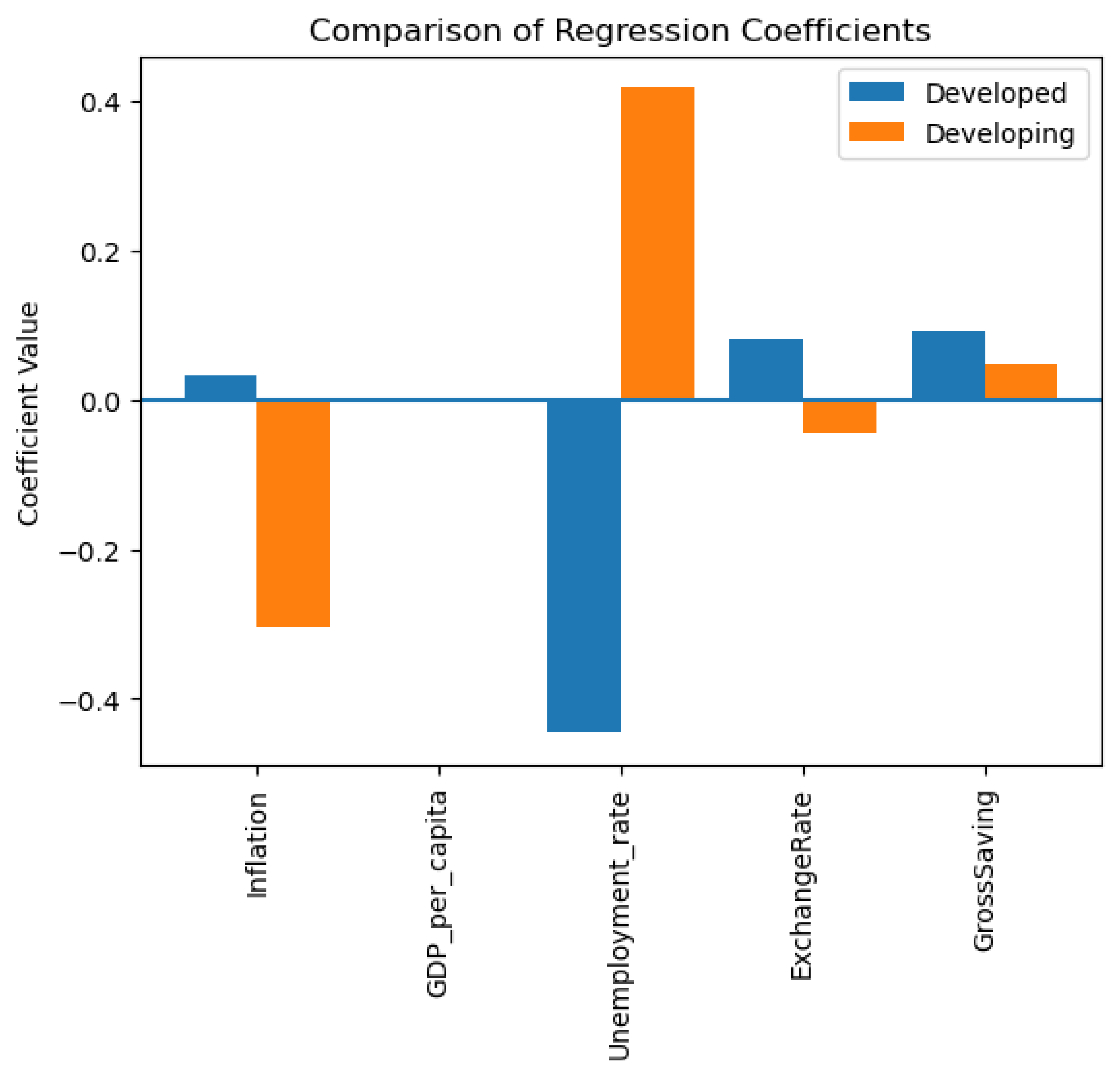
Variable	Correlation with Consumption
GDP per capita	0.29
Inflation	0.27
GrossSaving	0.07
ExchangeRate	-0.15
Year	-0.20
Unemployment_rate	-0.49

Correlation with Consumption – Developing Countries

	Variable	Correlation with Consumption
0	GDP_per_capita	0.23
1	GrossSaving	0.13
2	Unemployment_rate	0.12
3		
	Inflation	-0.23
4		
5	ExchangeRate	-0.24

Regression results: Coefficients Comparison

- INFLATION AFFECTS THE TWO GROUPS VERY DIFFERENTLY:
  - **Developed: +0.03 → little to no impact on consumption**
  - **Developing: –0.30 → higher inflation reduces consumption**
- THIS SUGGESTS CONSUMPTION IS MORE SENSITIVE TO INFLATION IN DEVELOPING COUNTRIES



- **MODEL FIT:**
- **DEVELOPED COUNTRIES R^2: 0.29**
- **DEVELOPING COUNTRIES R^2: 0.12**

Prediction Model Performance

MODEL SETUP

- TRAIN: 1991 – 2015
- TEST: 2015 – 2019
- MODEL: LINEAR REGRESSION
- FEATURES=["INFLATION","GDP_PER_CAPITA","UNEMPLOYMENT_RATE","EXCHANGERATE","GROSSSAVING"]
- TARGET = "CONSUMPTION"

Developed Countries:
- MAE: 0.80
- RMSE: 0.97
- R^2: **-1.37**

Developing Countries:
- MAE: 1.67
- RMSE: 1.87
- R^2: **-0.26**

Model Limitations & Next Step

- The linear regression model has **low explanatory power (low R^2)**
- This results suggest the model does not fully capture consumption patterns
- This may be because economic effects take **time to appear**
- To address this, I extend the model by including **lagged variables**

Lag Model Results (Lag1 vs Lag2)

Lag1 Model

$$Consumption_t = \beta_0 + \beta_1 Inflation_{t-1} + \beta_2 GDP_{t-1} + \beta_3 Unemployment_{t-1} + \beta_4 ExchangeRate_{t-1} + \beta_5 GrossSaving_{t-1}$$

```
=== Developed Countries ===
MAE : 1.08
RMSE: 1.2
R2  :-2.62
```

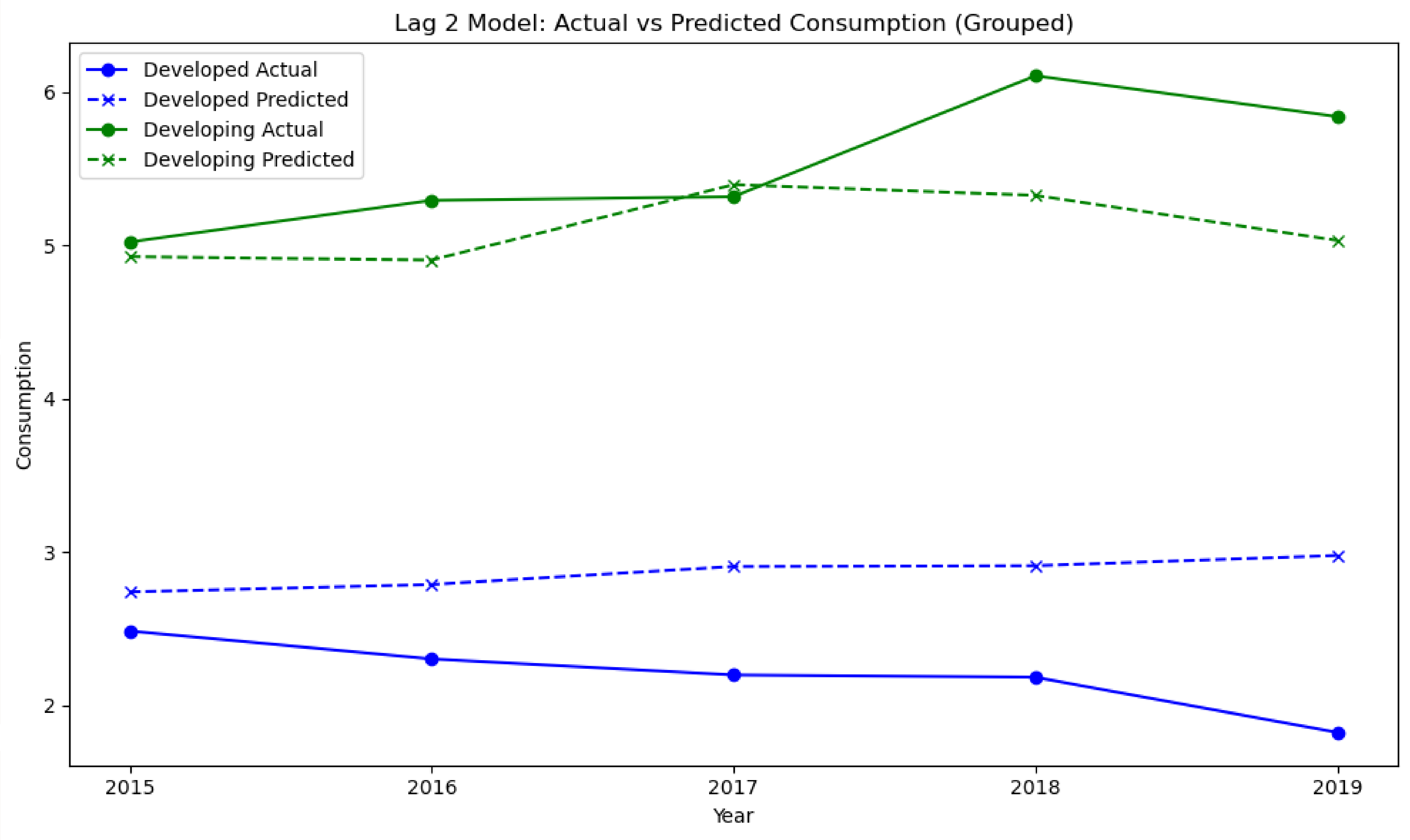
```
=== Developing Countries ===
MAE : 0.96
RMSE: 1.24
R2  : 0.45
```

Lag2 Model

$$Consumption_t = \beta_0 + \beta_1 Inflation_{t-2} + \beta_2 GDP_{t-2} + \beta_3 Unemployment_{t-2} + \beta_4 ExchangeRate_{t-2} + \beta_5 GrossSaving_{t-2}$$

```
=== Developed Countries ===
MAE : 0.74
RMSE: 0.82
R2  : -0.7
```

```
=== Developing Countries ===
MAE : 0.72
RMSE: 0.87
R2  : 0.73
```



Lag 2 Model: Actual vs Predicted Consumption (Grouped)
Developed Actual
Developed Predicted
Developing Actual
Developing Predicted
Consumption
2
3
4
5
6
2015
2016
2017
2018
2019
Year

Model Limitations (After Lag Model)

- The model improves with lagged variables, especially for developing countries
- However, autocorrelation still exists (low DW value)
- This suggests the model does not fully capture how past values affect current consumption
- Additional variables may be needed to better explain consumption
- Further improvement: include more lagged variables or use more advanced models

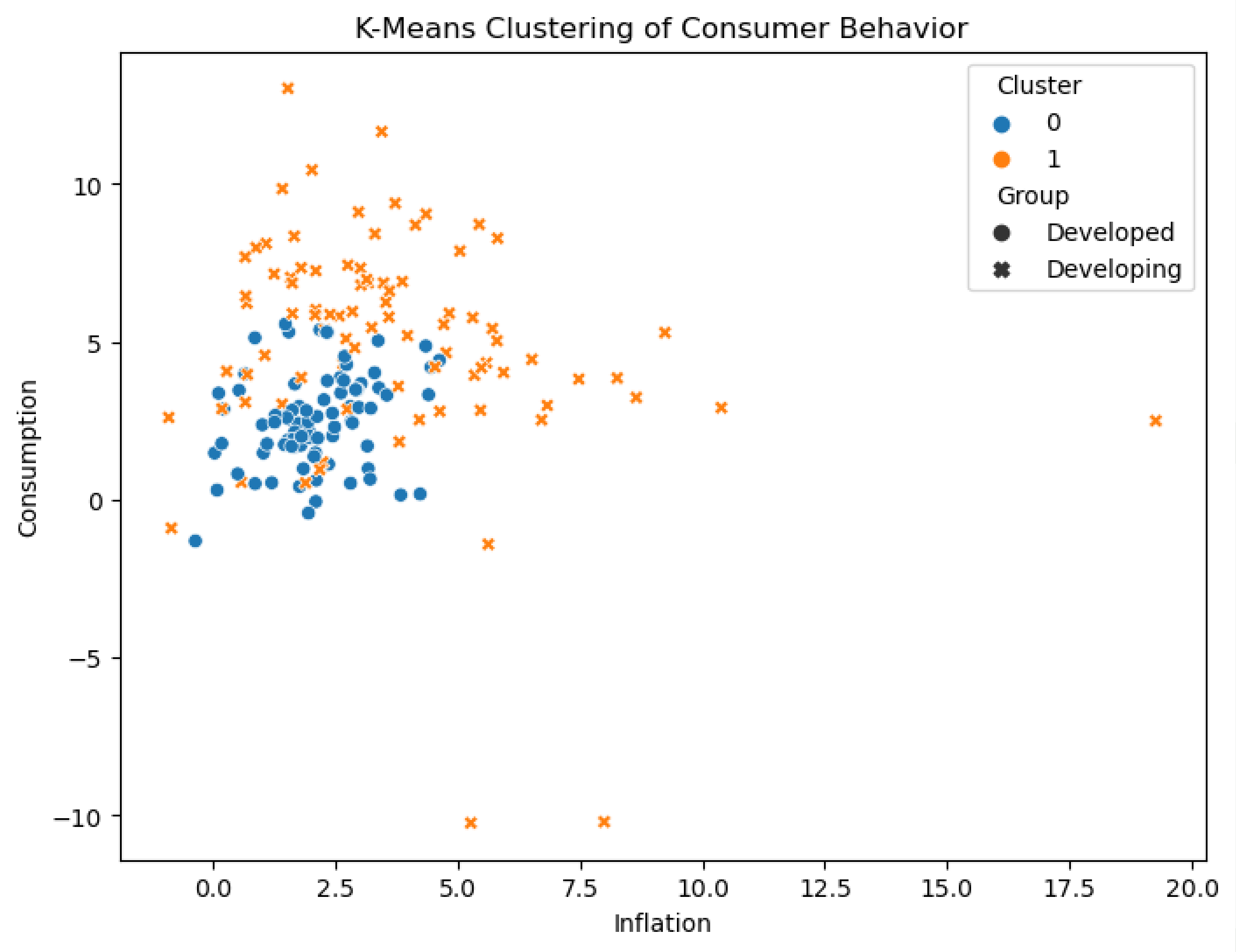
K-Means Clustering Results

Method

- K-Means Clustering (k = 2)
- Variables: Inflation, GDP, Unemployment, Exchange Rate, Gross Saving

Results

- Cluster 0: Mostly **Developed countries**
- Cluster 1: Mostly **Developing countries**



- Data naturally separates into two groups
- Strong structural differences between economies

Conclusion

- In developing countries, lagged variables (especially 2-year lags) significantly improve the model and better explain consumption, although some autocorrelation remains
- In developed countries, consumption is less predictable using these variables, suggesting the influence of additional factors
- Inflation has little effect in developed countries but reduces consumption in developing countries
- Overall, consumption behaviour differs across groups and responds to economic conditions with a delay
- These results suggest that consumption is more sensitive to economic changes like inflation in developing countries